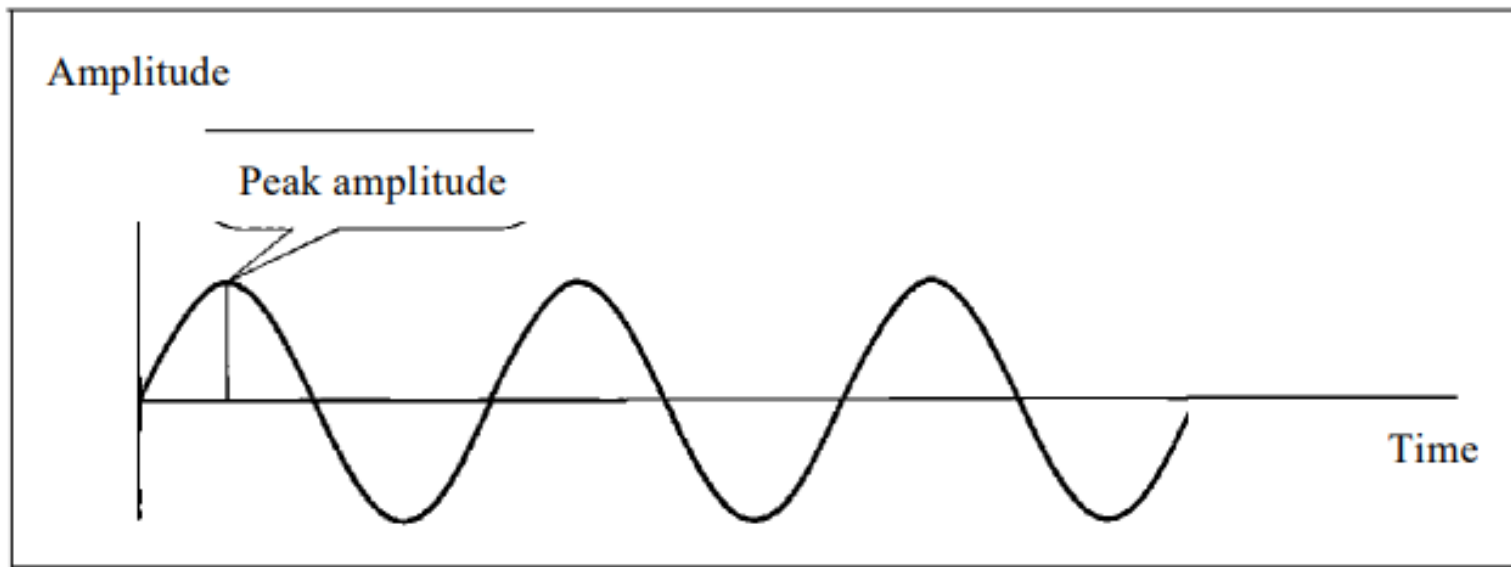


Physical Layer

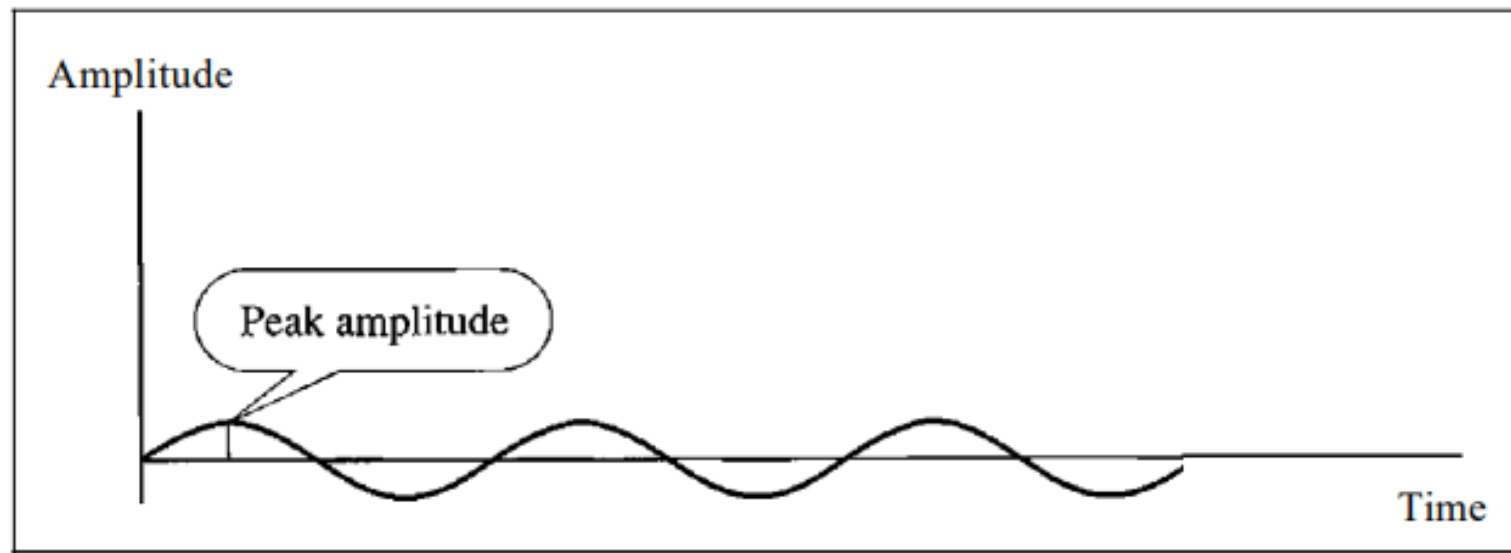
- **Its major functionality is to move data in the form of electromagnetic signals across transmission media.**
- **Data need to be change to signals before transmission.**

Analog & Digital Signal

- **Signals can be analog or digital.**
- **Analog data are continuous and take continuous values. Digital data have discrete states and take discrete values.**
- **Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values.**
- **Both analog and digital signals can take one of two forms: periodic or nonperiodic.**
- **A periodic signal completes a pattern within a measurable time frame, called a period and repeats that pattern over subsequent identical periods.**
- **The completion of one full pattern is called a cycle. A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.**



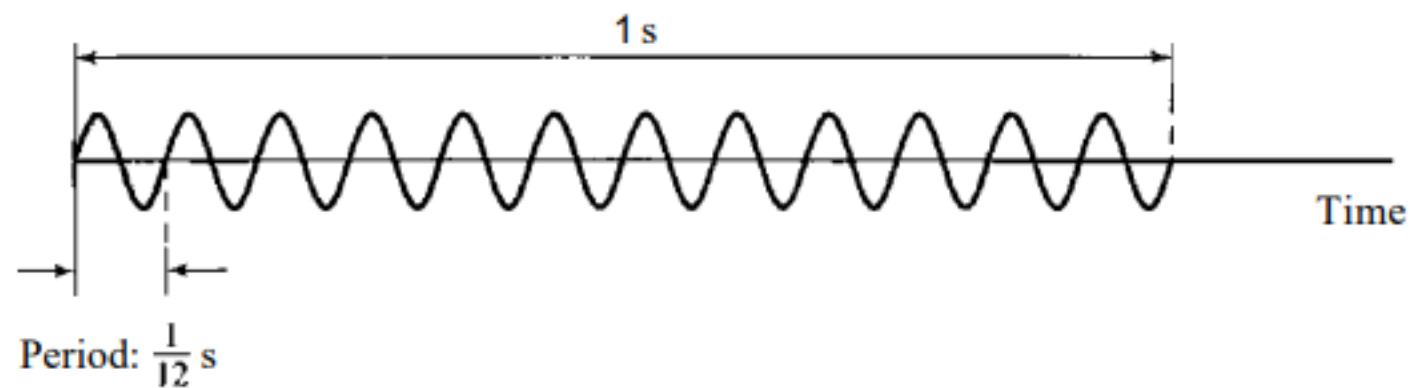
a. A signal with high peak amplitude



b. A signal with low peak amplitude

Amplitude

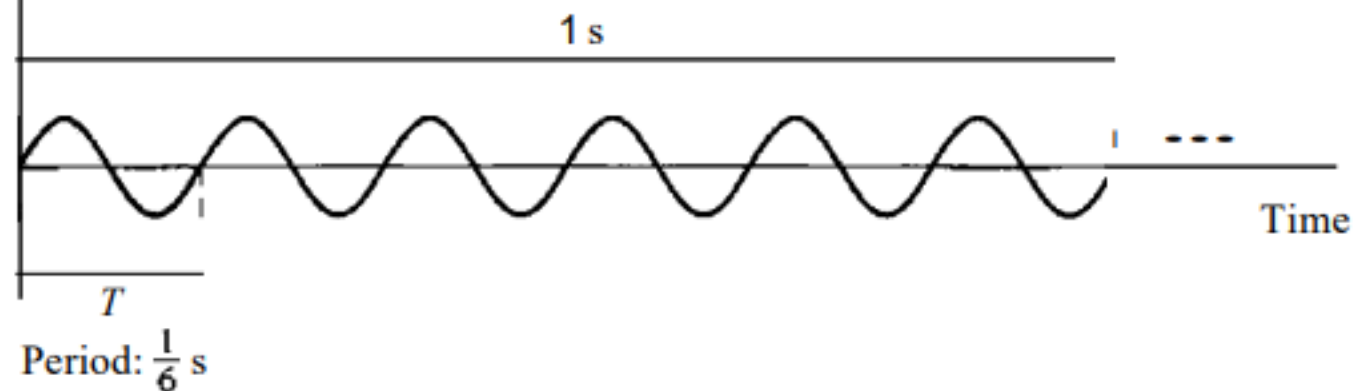
12 periods in 1 s $\longrightarrow$ Frequency is 12 Hz



a. A signal with a frequency of 12 Hz

Amplitude

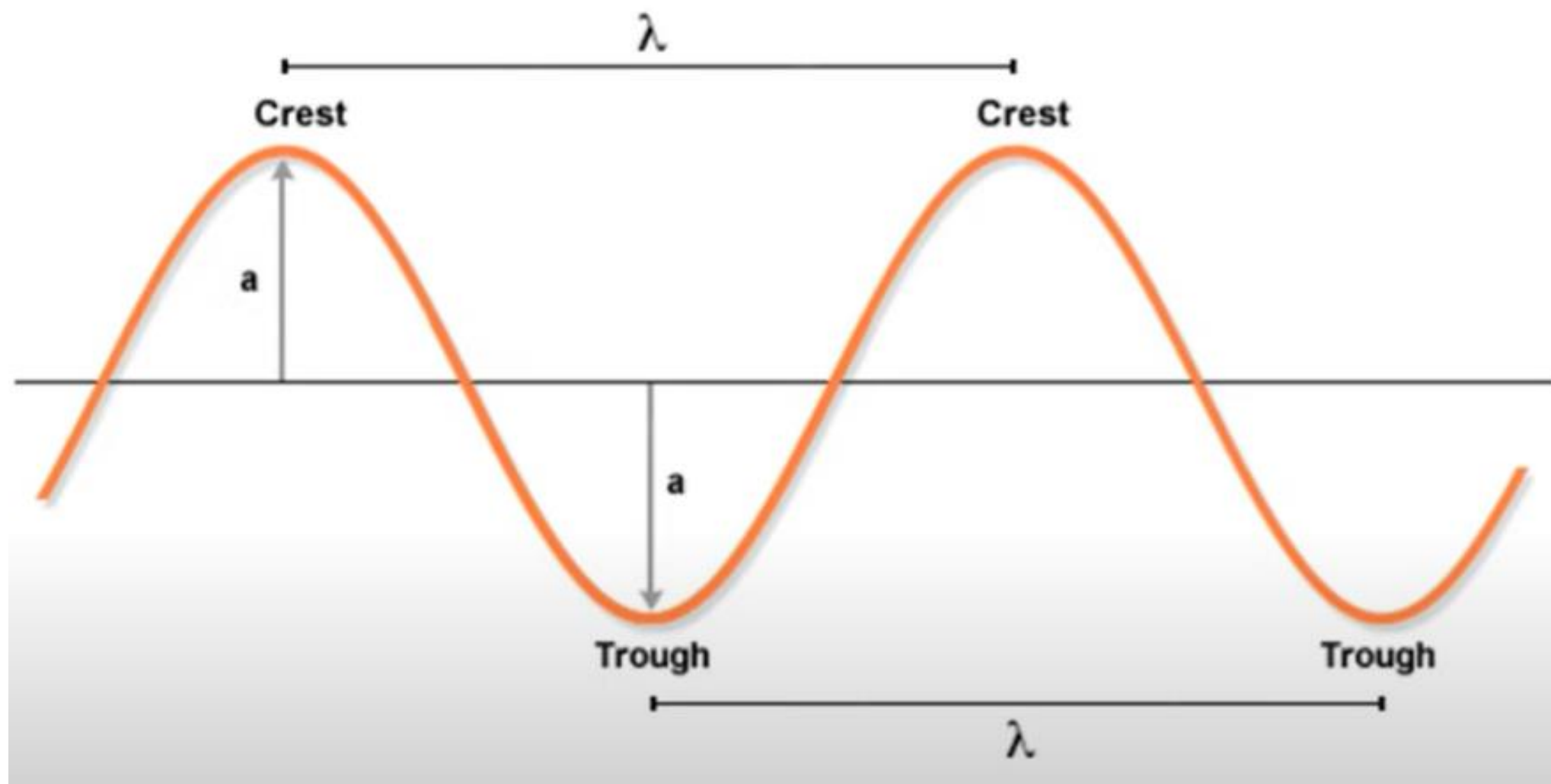
6 periods in 1 s $\longrightarrow$ Frequency is 6 Hz

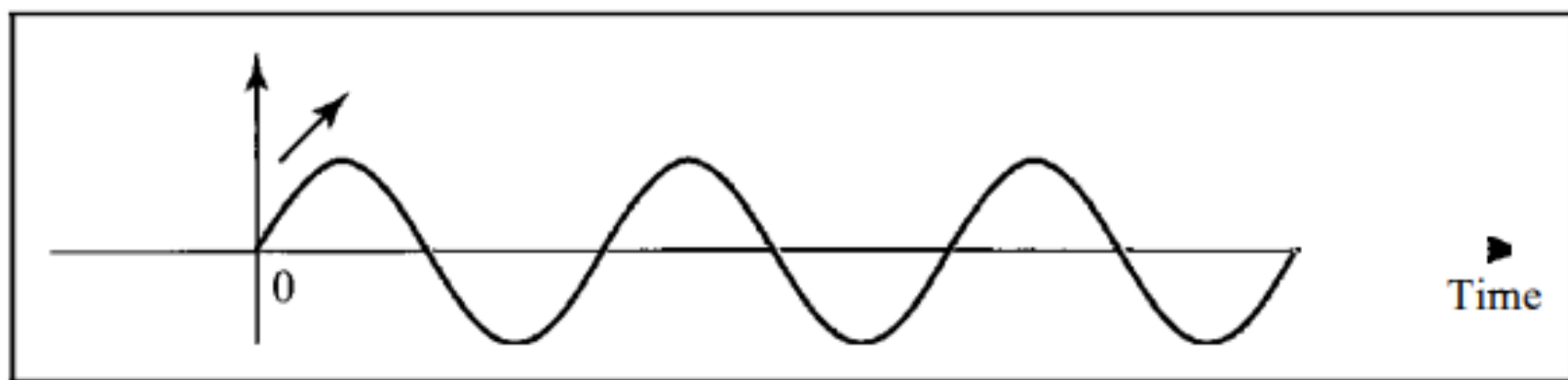


b. A signal with a frequency of 6 Hz

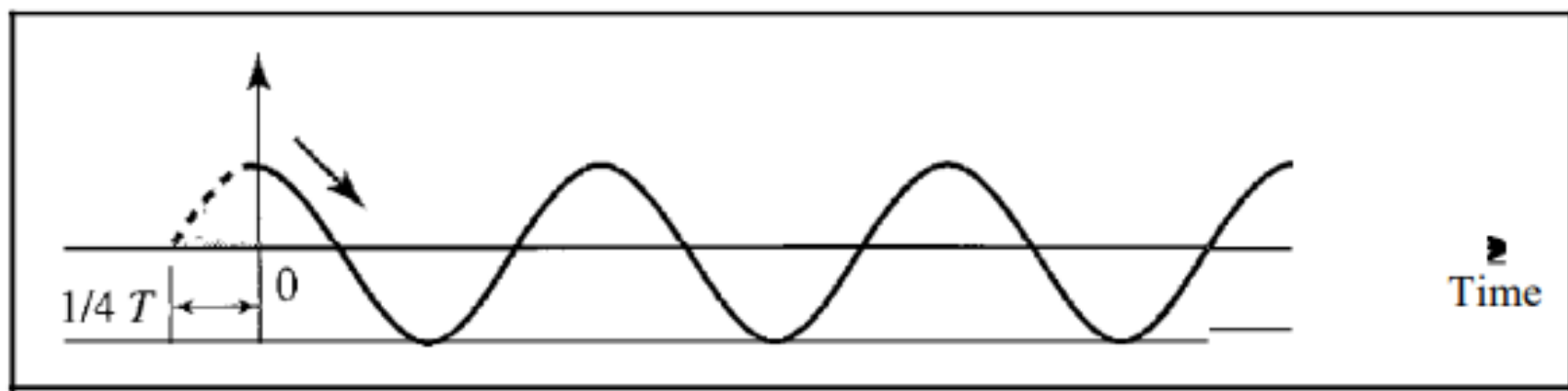
Table 3.1 *Units of period and frequency*

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz





a. 0 degrees



b. 90 degrees